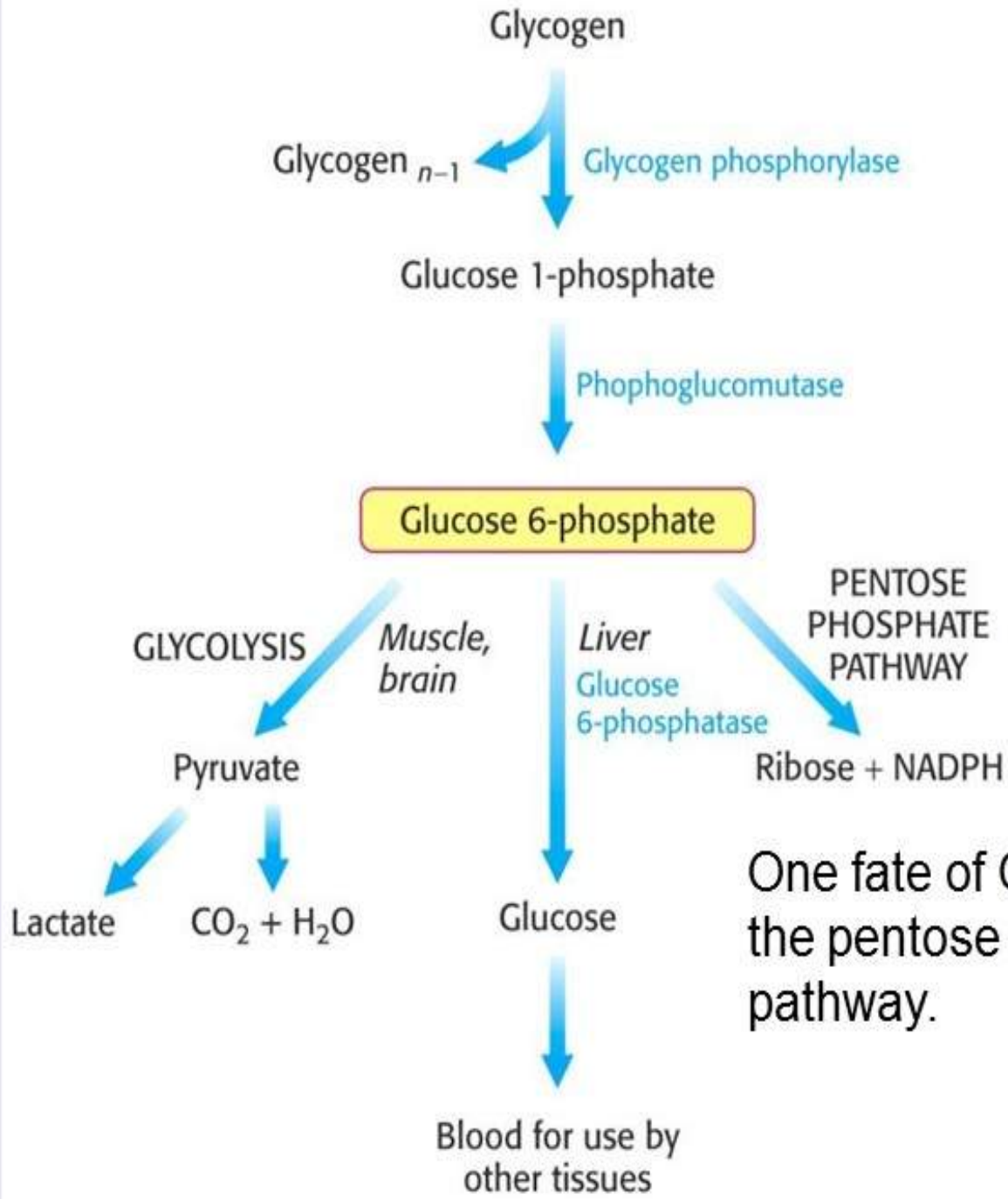
HMP Shunt

Pentose Phosphate Pathway

Also known as:

- Pentose shunt
- Hexose monophosphate shunt
- Phosphogluconate pathway

- It occurs in the cytosol.

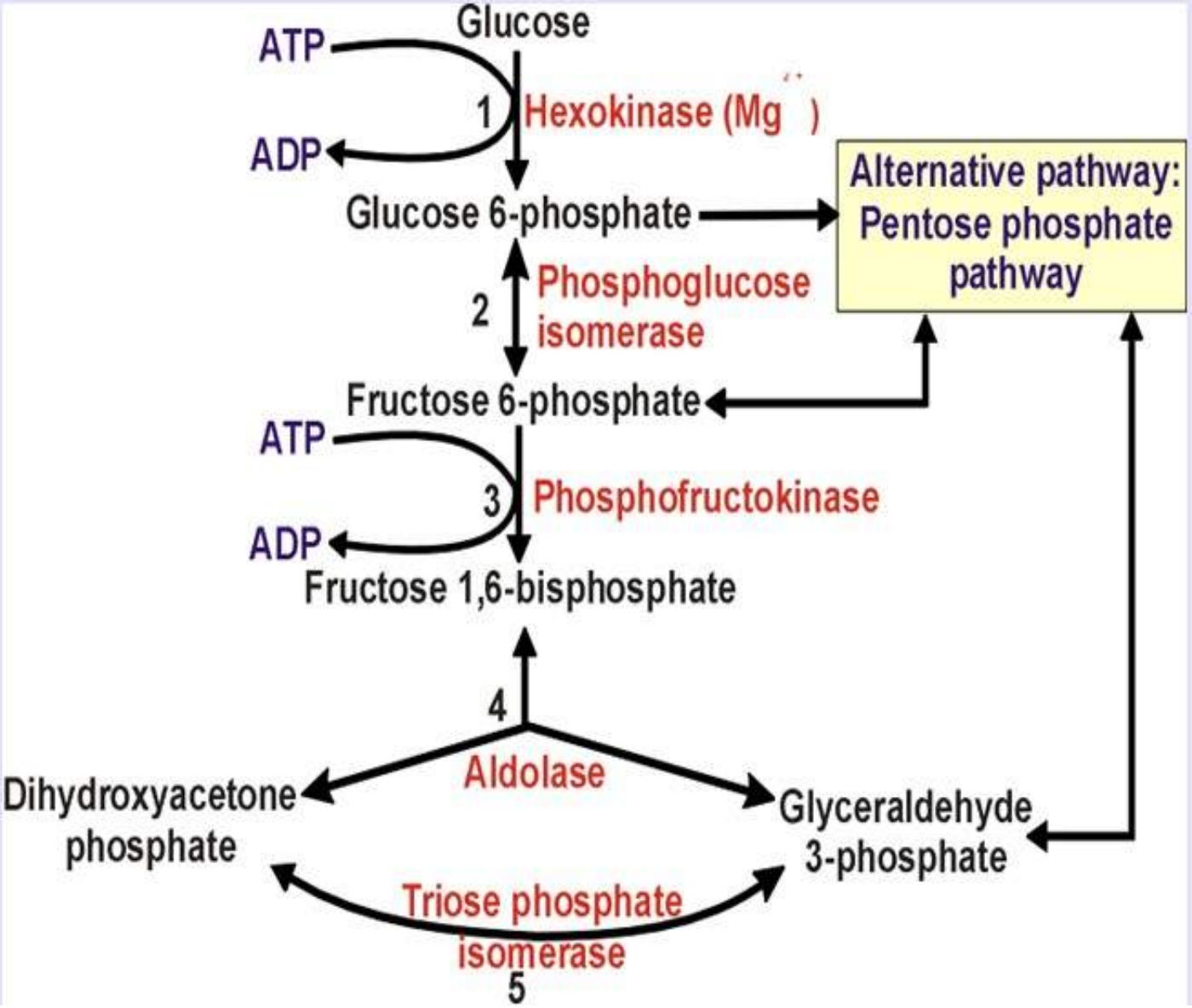


One fate of G6P is the pentose pathway.

The pentose pathway is a shunt.

- The pathway begins with the glycolytic intermediate glucose 6-P.
- It reconnects with glycolysis because two of the end products of the pentose pathway are glyceraldehyde 3-P and fructose 6-P; two intermediates further down in the glycolytic pathway.
- It is for this reason that the pentose pathway is often referred to as a shunt.

It's a shunt


Glucose
ATP
1
Hexokinase (Mg++)
ADP
Glucose 6-phosphate
Alternative pathway: Pentose phosphate pathway
2
Phosphoglucose isomerase
Fructose 6-phosphate
ATP
3
Phosphofructokinase
ADP
Fructose 1,6-bisphosphate
4
Aldolase
Dihydroxyacetone phosphate
Glyceraldehyde 3-phosphate
Triose phosphate isomerase
5

What does the pentose phosphate pathway achieve?

- The pathway yields reducing potential in the form of NADPH to be used in anabolic reactions requiring electrons.
- The pathway yields ribose 5-phosphate.
  - Nucleotide biosynthesis leading to:
    - DNA
    - RNA
    - Various cofactors (CoA, FAD, SAM, NAD^+/$NADP^+$).

Significance of NADPH+H$^+$

Synthesis

Fatty acid biosynthesis

Cholesterol biosynthesis

Neurotransmitter biosynthesis

Nucleotide biosynthesis

Detoxification

Reduction of oxidized glutathione

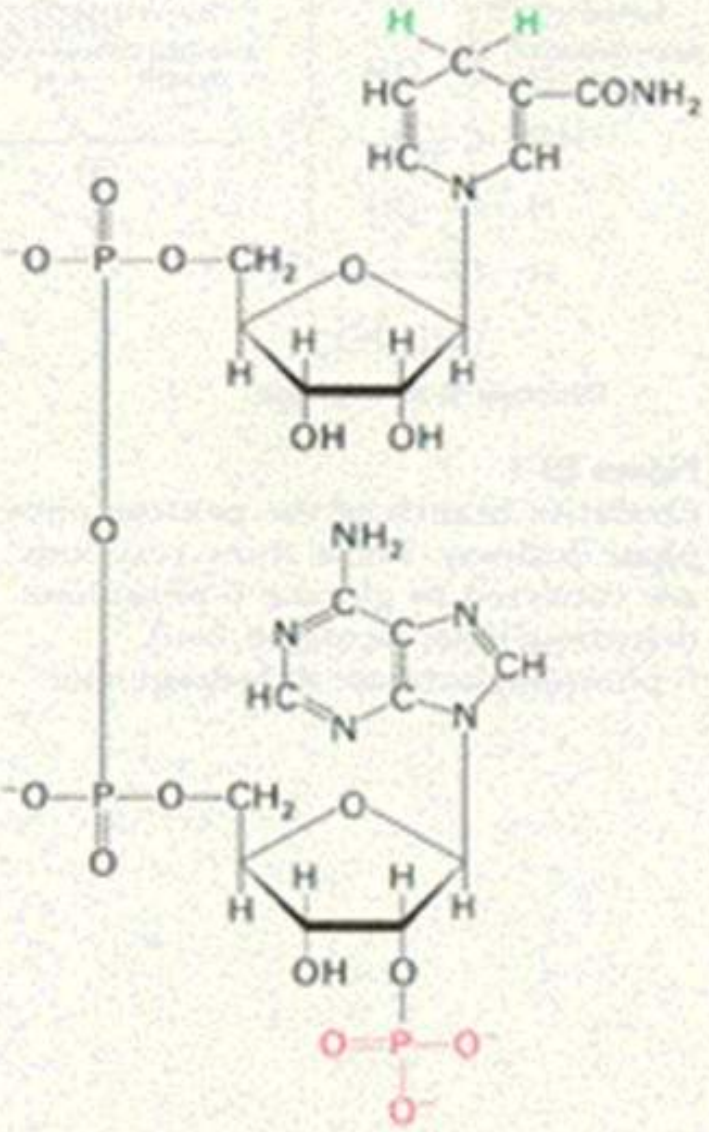
Cytochrome P450 monooxygenases

Sites of HMP Shunt

Tissue	Function
Adrenal gland	Steroid synthesis
Liver	Fatty acid and cholesterol synthesis
Testes	Steroid synthesis
Adipose tissue	Fatty acid synthesis
Ovary	Steroid synthesis
Mammary gland	Fatty acid synthesis
Red blood cells	Maintenance of reduced glutathione

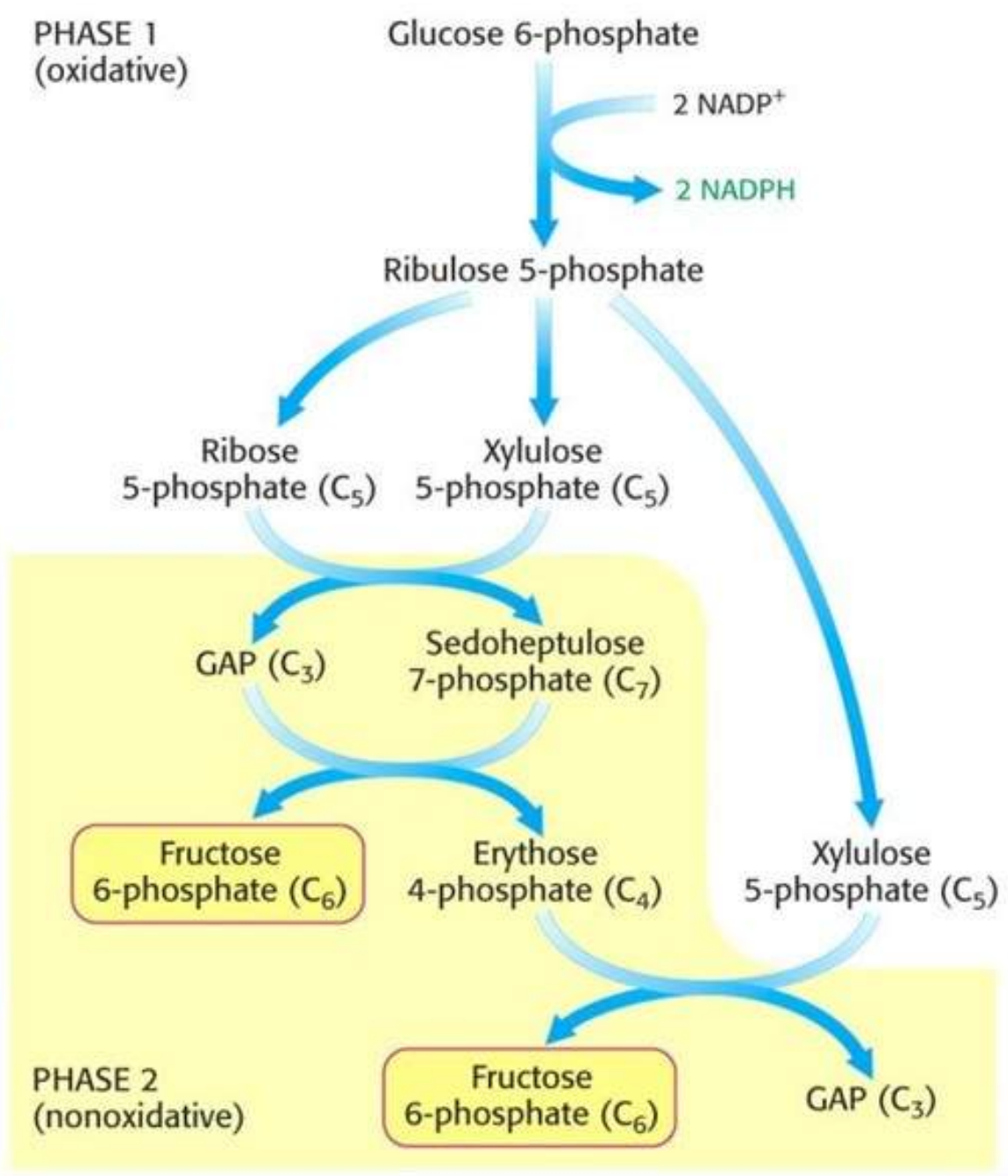
NADPH is a phosphorylated form of NADH.

In general, with some exceptions, NADH is used to drive the phosphorylation of ADP to ATP. NADPH is used where reducing potential is required for synthetic reactions.



Reduced nicotinamide adenine dinucleotide phosphate (NADPH)

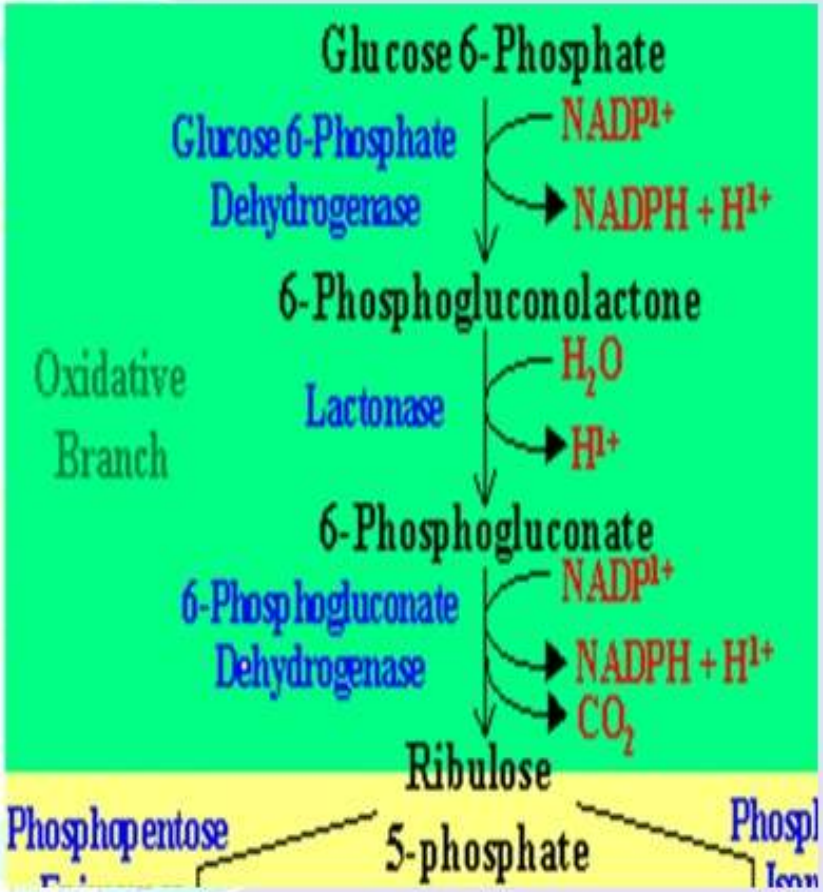
The pentose pathway can be divided into two phases.



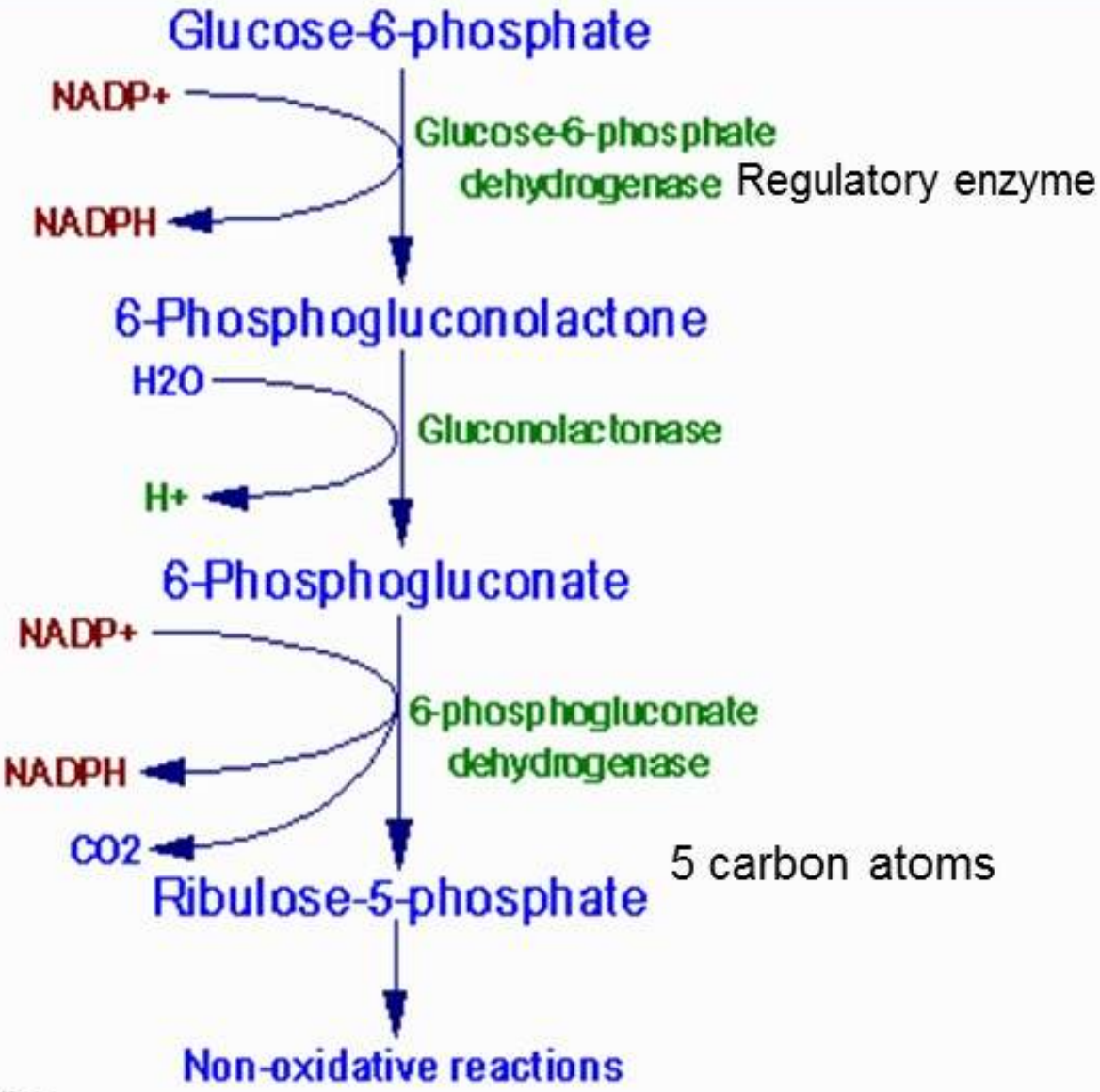
Non-oxidative interconversion of sugars

Oxidative Phase

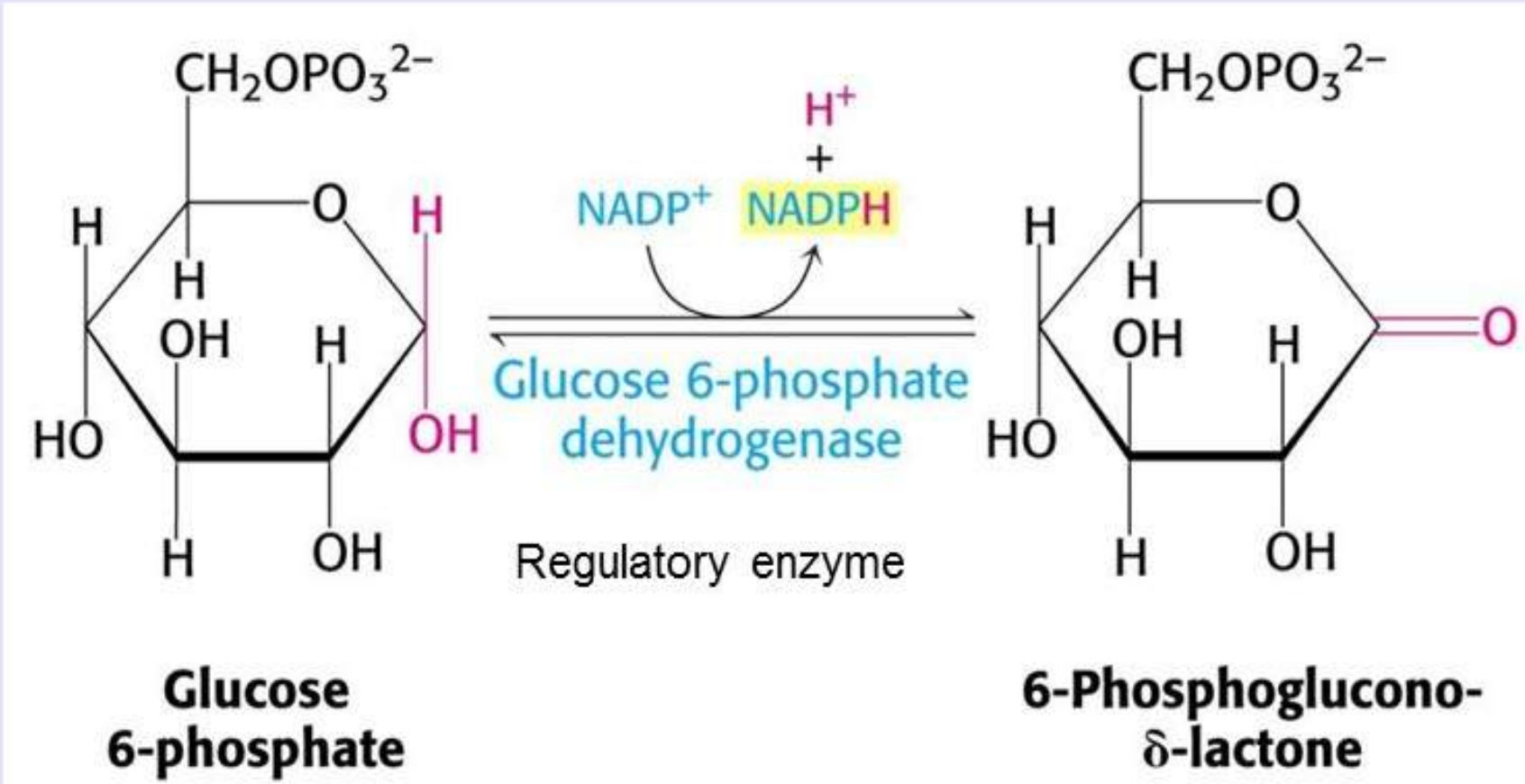
- NADPH + H^+ is formed from two separate reactions.
- The glucose 6-phosphate DH (G6PD) reaction is the rate limiting step and is essentially irreversible.
- Cells have a greater need for NADPH than ribose 5-phosphate.



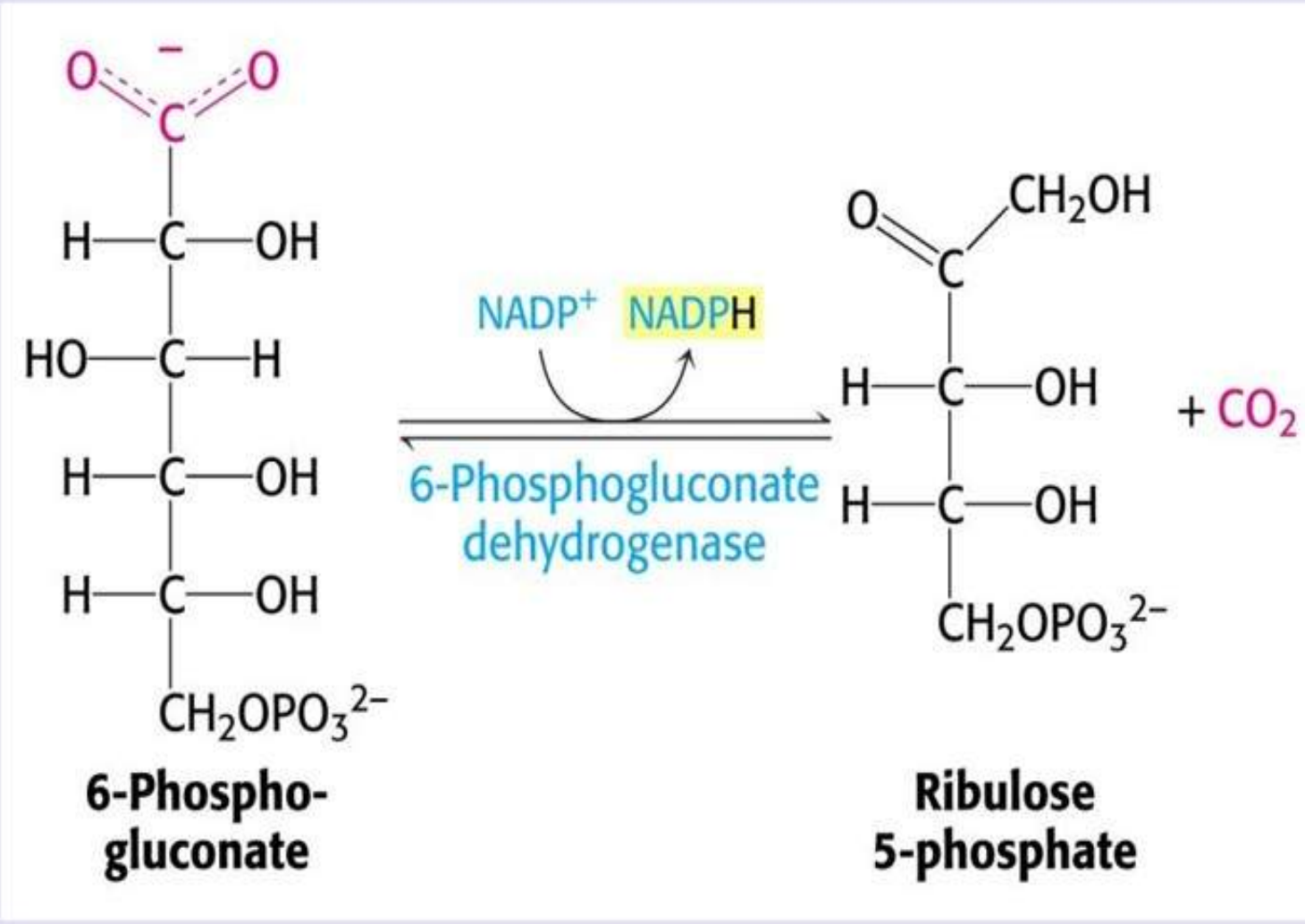
Oxidative Stage of Pentose Phosphate Pathway



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The enzyme is highly specific for $NADP^+$; the K_m for NAD^+ is 1000 greater than for $NADP^+$.

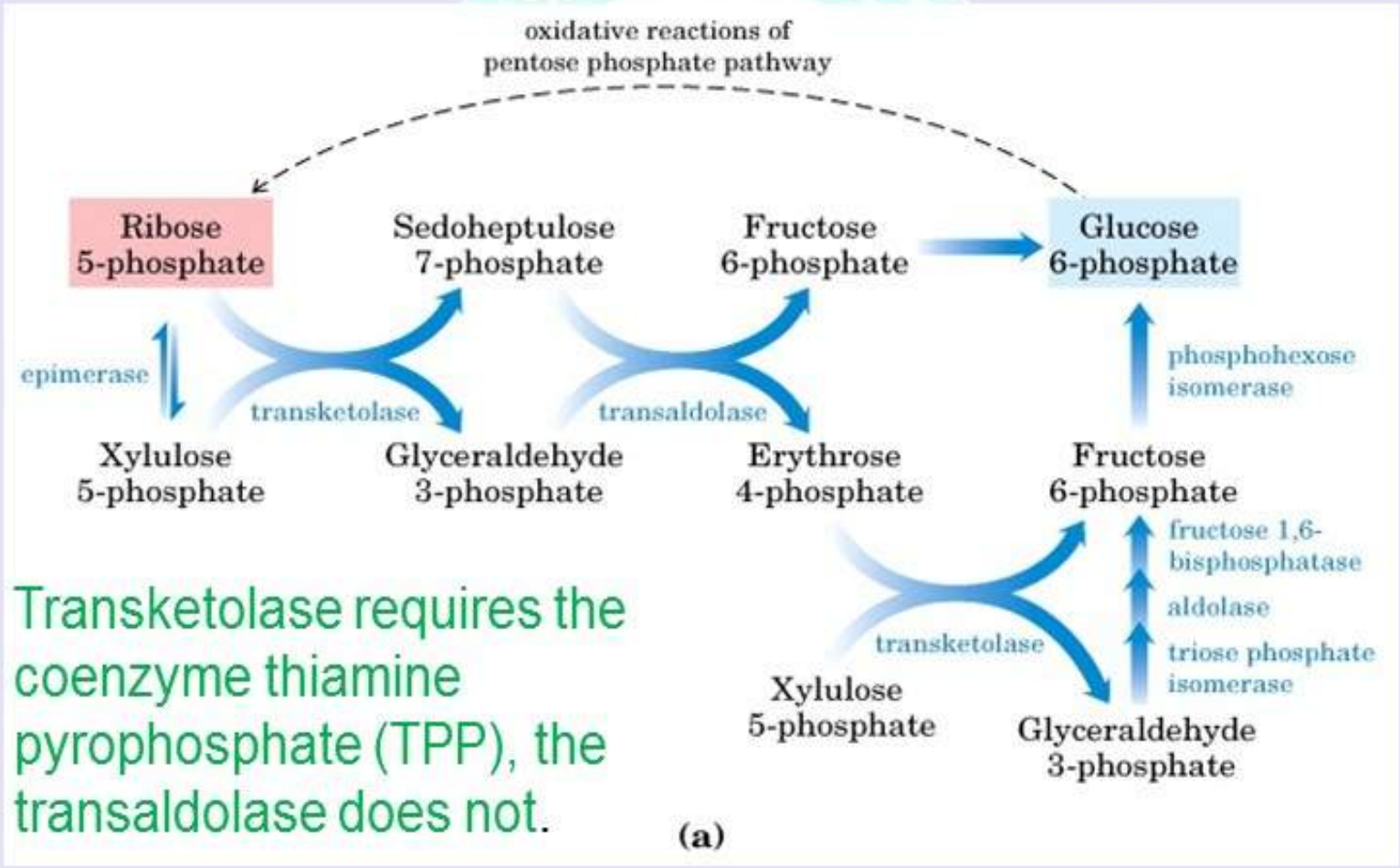

O
C
O
H—C—OH
HO—C—H
H—C—OH
H—C—OH
CH2OPO3 2–
NADP+
NADPH
6-Phosphogluconate dehydrogenase
O
CH2OH
C
H—C—OH
H—C—OH
CH2OPO3 2–
+ CO2
6-Phospho-gluconate
Ribulose 5-phosphate

Hexose Monophosphate Shunt Reactions

Reaction	Enzyme
Oxidative phase	
Glucose 6-phosphate + $NADP^+$ $\longrightarrow$ 6-phosphoglucono-δ-lactone + NADPH + H^+	Glucose 6-phosphate dehydrogenase
6-Phosphoglucono-δ-lactone + H_2O $\longrightarrow$ 6-phosphogluconate + H^+	Lactonase
6-Phosphogluconate + $NADP^+$ $\longrightarrow$ ribulose 5-phosphate + CO_2 + NADPH	6-Phosphogluconate dehydrogenase
Nonoxidative Phase	
Ribulose 5-phosphate $\rightleftharpoons$ ribose 5-phosphate	Phosphopentose isomerase
Ribulose 5-phosphate $\rightleftharpoons$ xylulose 5-phosphate	Phosphopentose epimerase
Xylulose 5-phosphate + ribose 5-phosphate $\rightleftharpoons$ sedoheptulose 7-phosphate + glyceraldehyde 3-phosphate	Transketolase
Sedoheptulose 7-phosphate + glyceraldehyde 3-phosphate $\rightleftharpoons$ fructose 6-phosphate + erythrose 4-phosphate	Transaldolase
Xylulose 5-phosphate + erythrose 4-phosphate $\rightleftharpoons$ fructose 6-phosphate + glyceraldehyde 3-phosphate	Transketolase

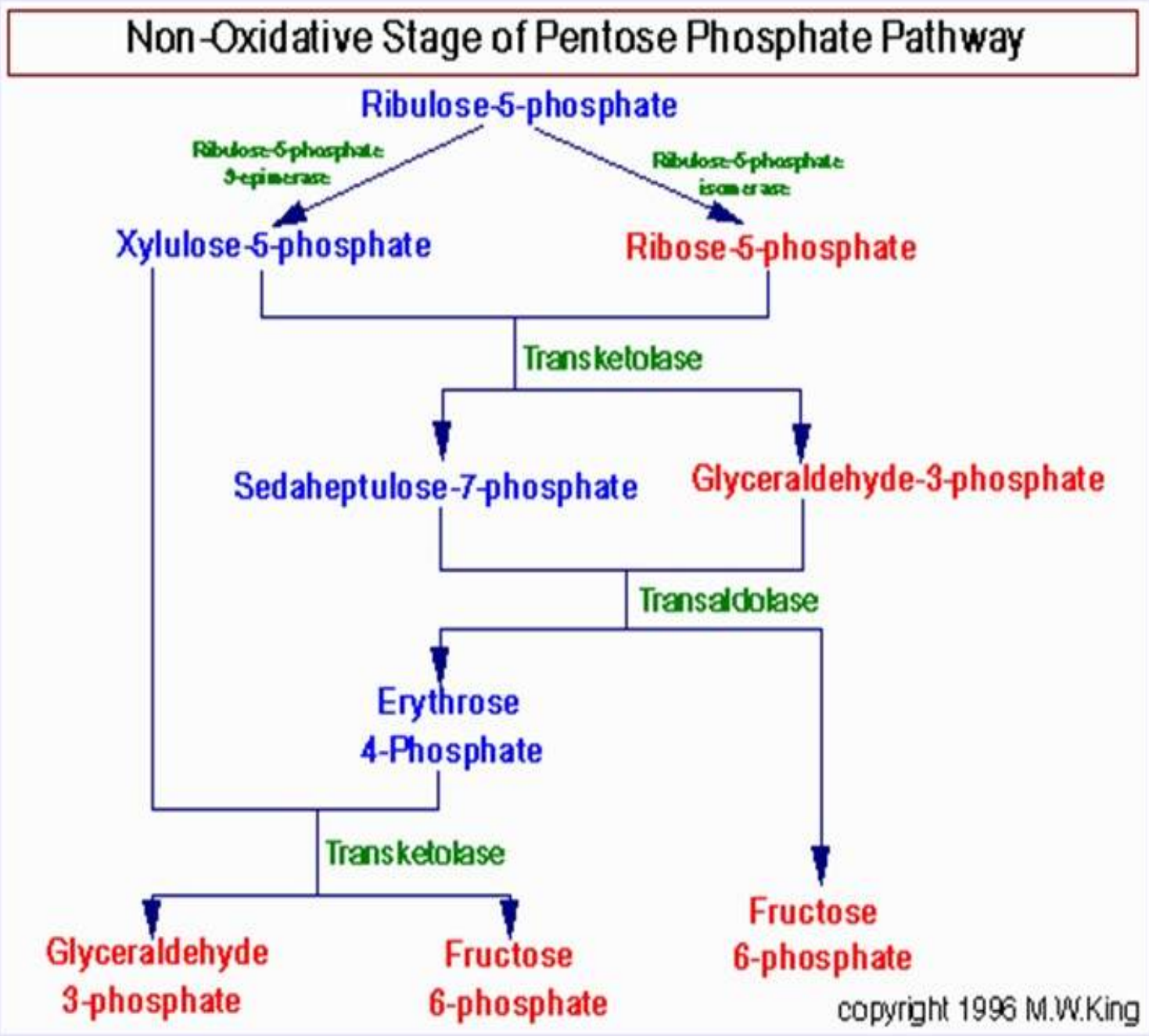
The nonoxidative phase of the pentose pathway

This entails extensive carbon atom rearrangement.

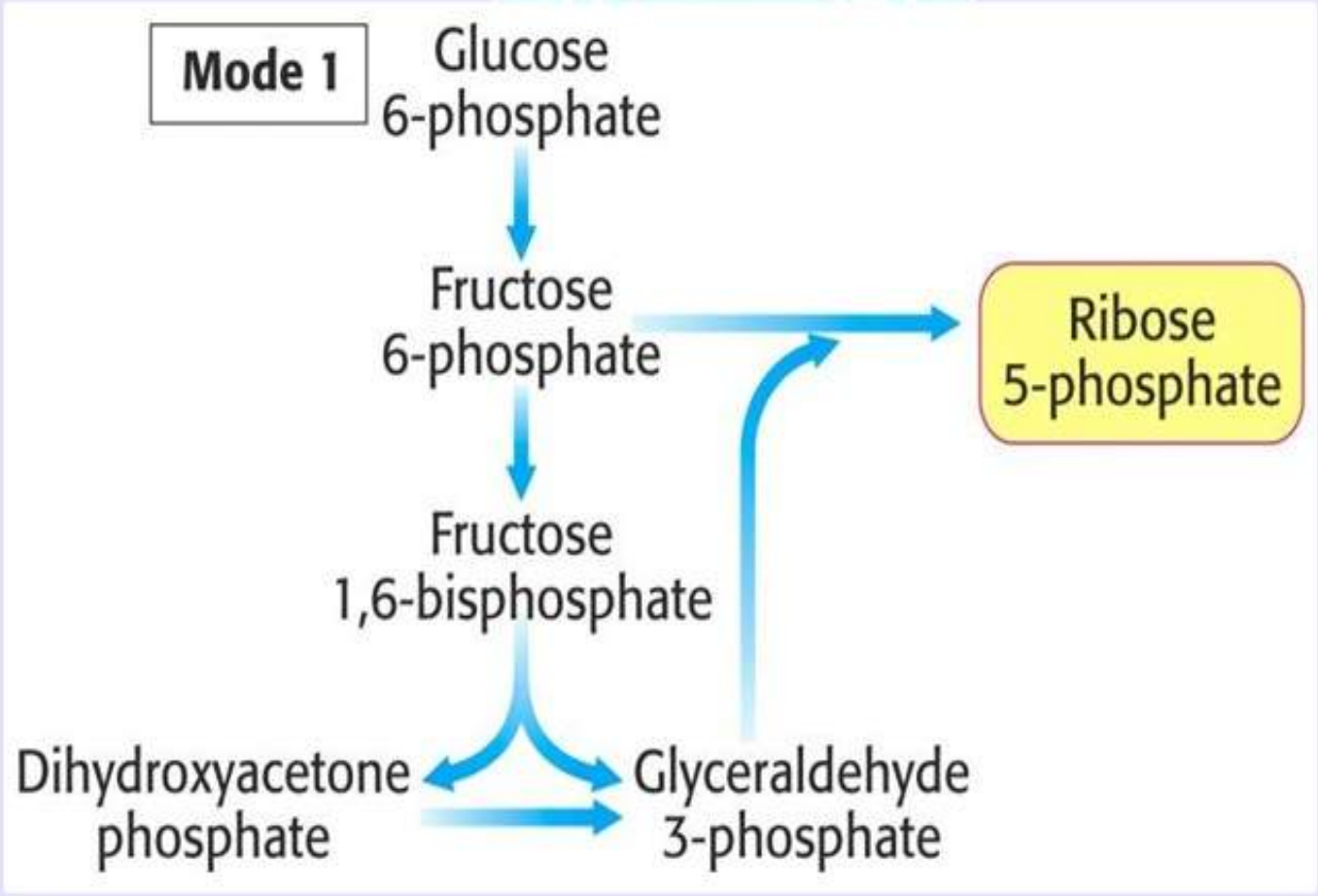


Transketolase requires the coenzyme thiamine pyrophosphate (TPP), the transaldolase does not.

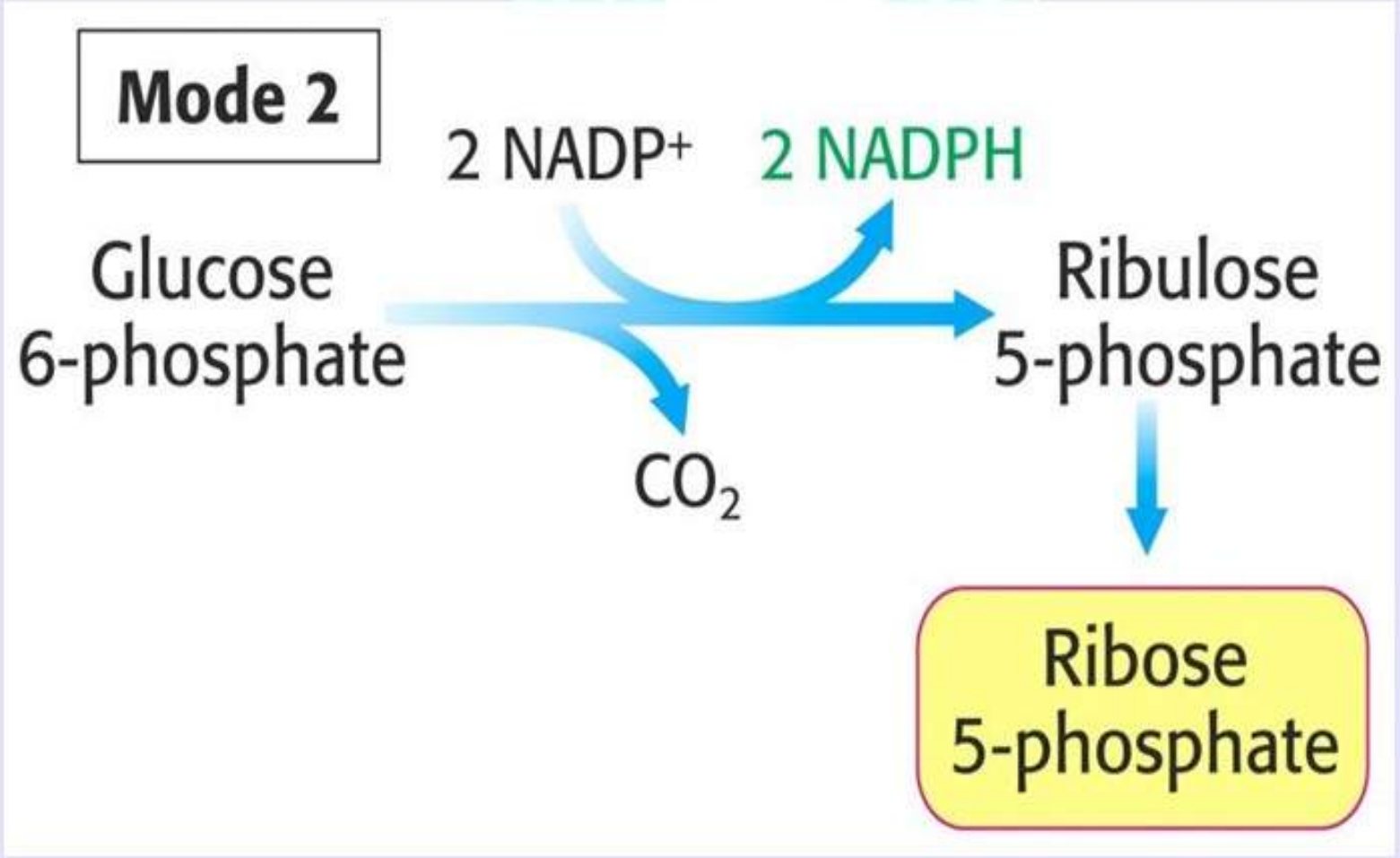
Non-Oxidative Stage of Pentose Phosphate Pathway



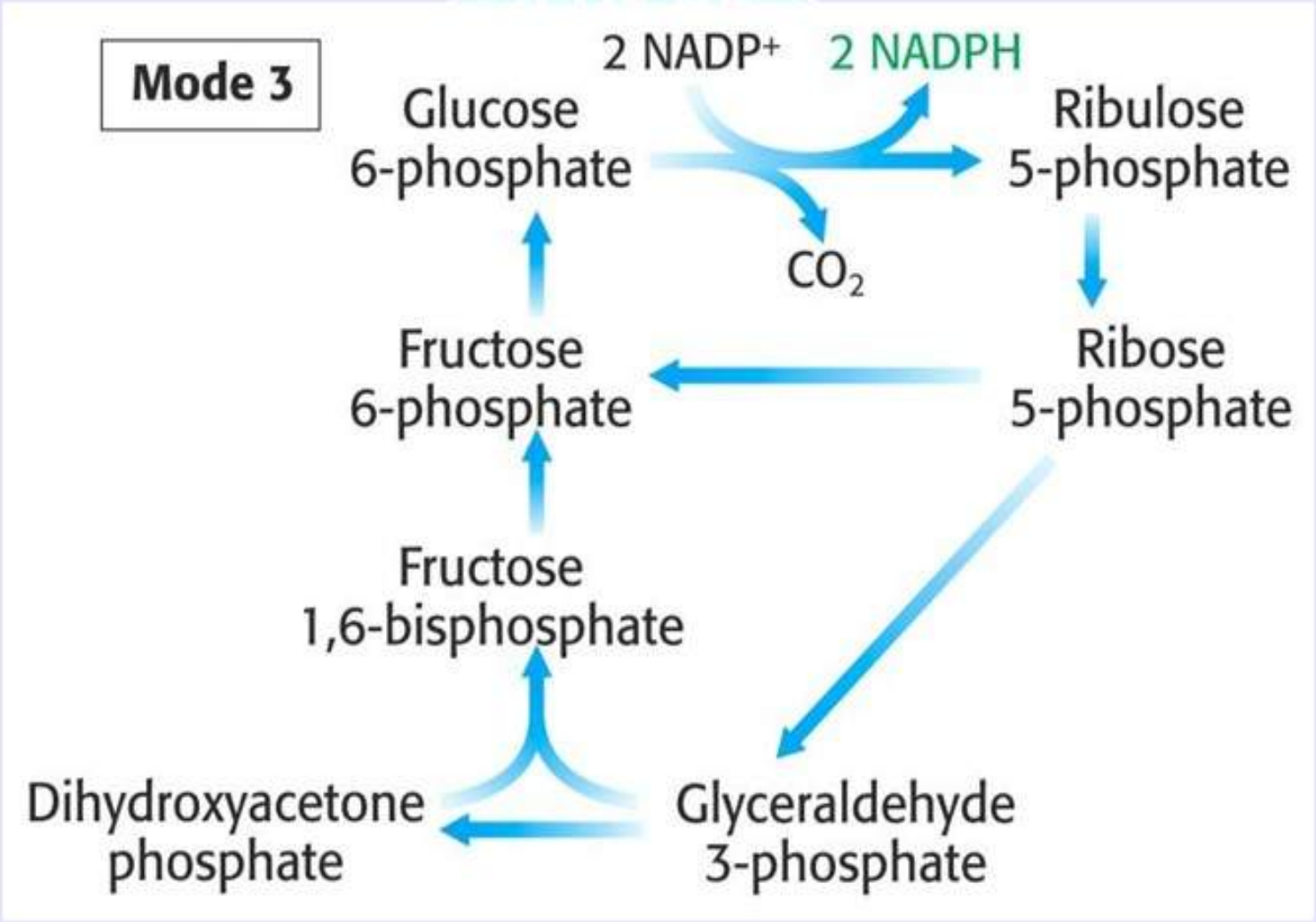
Rapidly dividing cells require more ribose 5-phosphate than NADPH.



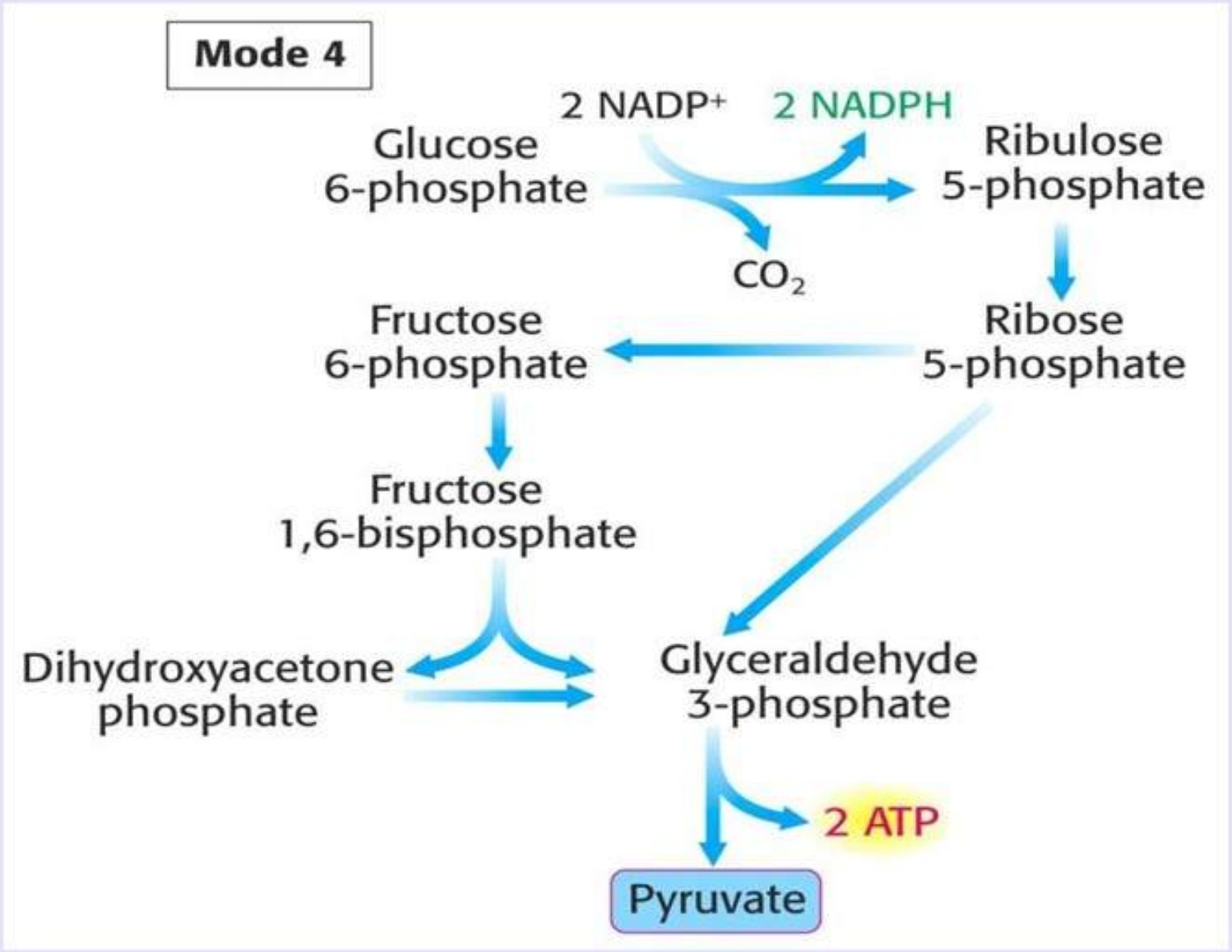
The need for NADPH and ribose 5-phosphate is balanced.



More NADPH is needed than ribose 5-phosphate;
Fatty acid synthesis in adipose cells.

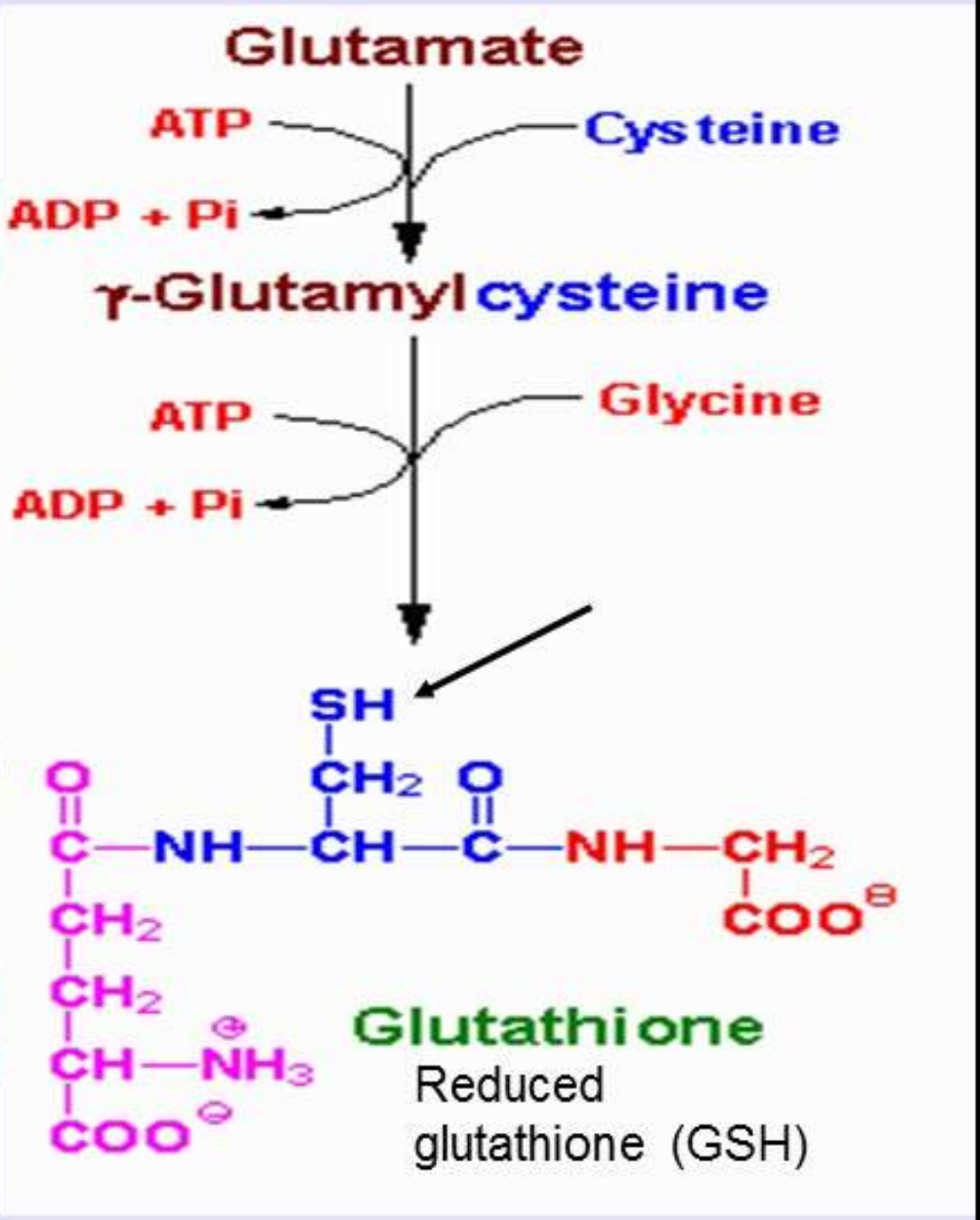


The cell needs both NADPH and ATP



Glutathione is a tri-peptide composed of glutamate, cystein, glycine.

Reduced glutathione (GSH) maintains the normal reduced state of the cell.

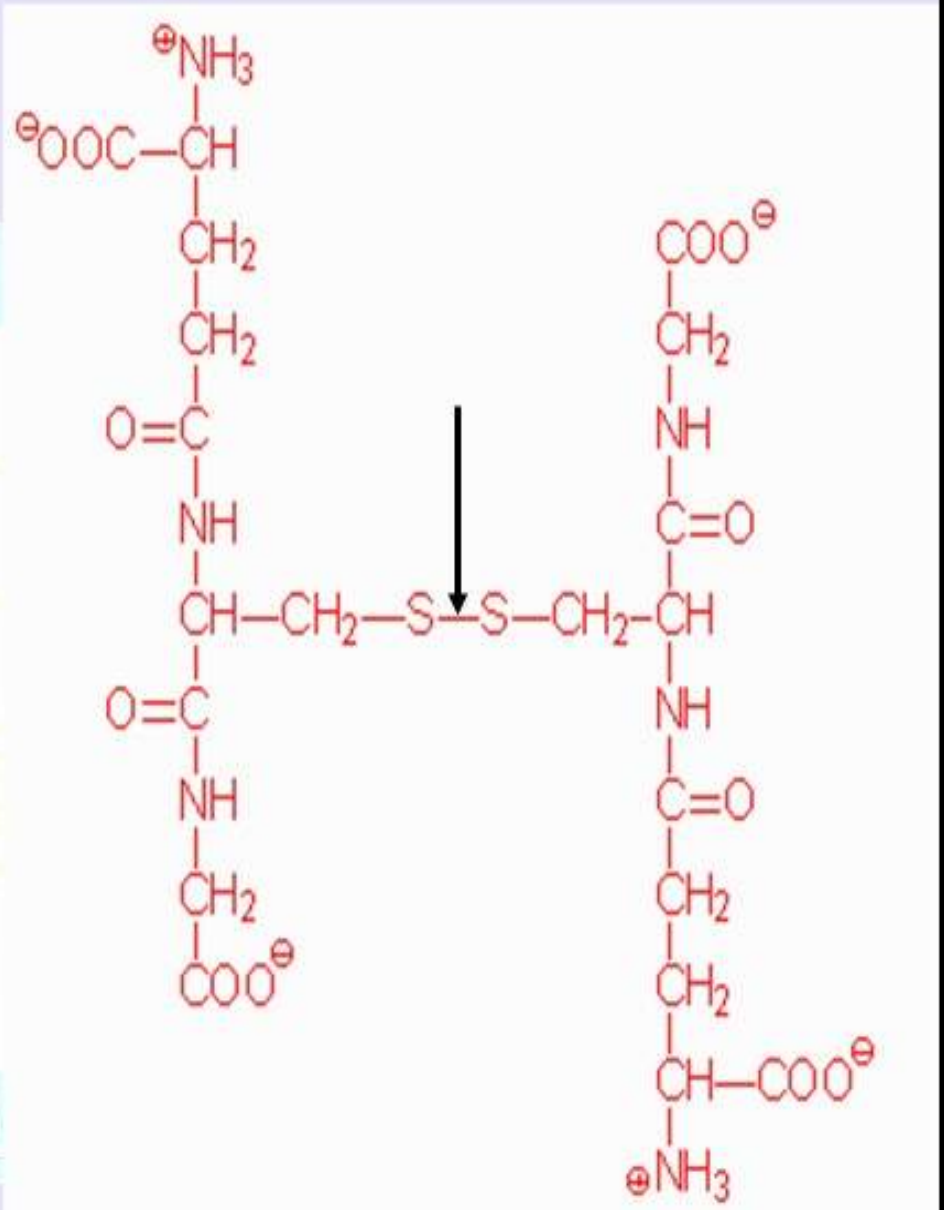


Glutathione Functions

- It serves as a reductant.
- Conjugates to drugs making them water soluble.
- Involved in amino acid transport across cell membranes.
- Cofactor in some enzymatic reactions.
- The sulfhydryl of GSH is used to reduce peroxides (ROS) formed during oxygen transport.
  - Reactive oxygen species (ROS) damage macromolecules (DNA, RNA, and protein) and ultimately lead to cell death.

The enzyme glutathione reductase uses NADPH as a cofactor to reduce GSSG back to two moles of GSH.

Thus, the pentose pathway is linked to the supply of adequate amounts of GSH.



Glutathione disulfide (GSSG)



Thank you